

PHYSICS 1

UNIT 4: FLUIDS, OSCILLATIONS & WAVES

Fall 2026

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Name: _____

How to Use This Packet

Each section below has two parts, in the Cornell note-taking format, plus a summary box.

Notes column (right, wide)

Fill this in during lecture. Definitions, formulas, and key terms are partly blanked — write the missing words, symbols, or numbers as your instructor presents them.

Cue column (left, narrow)

These are the key questions and terms for the section. Use them right after class (or before a quiz) to test yourself: cover the notes column and see if you can answer each cue from memory.

Summary box

At the end of each section, write 2–3 sentences in your own words capturing the main idea. If you can't summarize it, you don't know it yet — go back and review.

Chapter 11: Fluid Statics

11.2 Density

CUE / KEY QUESTIONS <i>Define density. Give the equation.</i> <i>SI unit of density?</i>	Density (ρ) is mass per unit volume: $\rho = \underline{\hspace{2cm}}$ SI unit of density: $\underline{\hspace{2cm}}$ (also commonly seen as g/cm^3 , where $1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$). Density lets you compare materials independent of how much of them you have — a small dense object can outweigh a much larger, less dense one.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

11.3 Pressure

CUE / KEY QUESTIONS <i>Define pressure. Give the equation.</i> <i>SI unit of pressure?</i>	Pressure (P) is force per unit area: $P = \underline{\hspace{2cm}}$ SI unit of pressure: the $\underline{\hspace{2cm}}$, where $1 \text{ Pa} = 1 \underline{\hspace{2cm}}$ For a given force, spreading it over a larger area $\underline{\hspace{2cm}}$ the pressure — this is why snowshoes keep you from sinking.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

11.4 Variation of Pressure with Depth in a Fluid

CUE / KEY QUESTIONS <i>Equation for pressure at depth h in a fluid?</i> <i>Does pressure at a given depth depend on the shape or amount of the container?</i>	Pressure due to the weight of a fluid at depth h : $P = \underline{\hspace{2cm}}$ (this is added on top of any pressure already present at the surface, e.g. atmospheric pressure). Total (absolute) pressure at depth h below a surface with pressure P_0 : $P = P_0 + \underline{\hspace{2cm}}$ Pressure at a given depth depends only on the fluid's $\underline{\hspace{2cm}}$ and the $\underline{\hspace{2cm}}$ itself — $\underline{\hspace{2cm}}$ on the shape or total amount of fluid present.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

11.5 Pascal's Principle

CUE / KEY QUESTIONS <i>State Pascal's Principle.</i> <i>Hydraulic lift equation relating F_1, A_1, F_2, A_2?</i>	Pascal's Principle: a change in pressure applied to an enclosed fluid is transmitted $\underline{\hspace{2cm}}$ to every point of the fluid and to the walls of its container. Hydraulic lift: $F_1 / A_1 = \underline{\hspace{2cm}}$ — a small force on a small-area piston can produce a much larger force on a large-area piston.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

11.6 Gauge Pressure, Absolute Pressure, and Pressure Measurement

CUE / KEY QUESTIONS	$\underline{\hspace{2cm}}$ pressure is the total actual pressure, measured relative to a true
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<i>Define gauge pressure vs. absolute pressure.</i> <i>Equation relating the two?</i>	vacuum (zero pressure). _____ pressure is measured relative to atmospheric pressure — it's what most pressure gauges (tire gauges, blood pressure cuffs) actually read. $P_{abs} = P_{gauge} + \underline{\hspace{2cm}}$
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

11.7 Archimedes' Principle

CUE / KEY QUESTIONS <i>State Archimedes' Principle.</i> <i>Equation for buoyant force?</i> <i>Floating vs. sinking — what determines which happens?</i>	Archimedes' Principle: the buoyant force on a submerged or floating object equals the _____. $F_b = \underline{\hspace{2cm}}$ An object floats if its average density is _____ the fluid's density; it sinks if its average density is _____ the fluid's density.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

Chapter 12: Fluid Dynamics

12.1 Flow Rate and Its Relation to Velocity

CUE / KEY QUESTIONS <i>Define flow rate. Give the equation and SI unit.</i> <i>State the equation of continuity, and what it means physically.</i>	Flow rate (Q) is the volume of fluid passing a point per unit time: $Q = \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$, SI unit: $\underline{\hspace{2cm}}$ Equation of continuity (for an incompressible fluid): $A_1 v_1 = \underline{\hspace{2cm}}$ Physically, continuity means flow rate is $\underline{\hspace{2cm}}$ along a pipe of varying cross-section — where the pipe narrows, the fluid speeds up, and where it widens, the fluid slows down.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

12.2 Bernoulli's Equation

CUE / KEY QUESTIONS <i>Write Bernoulli's Equation.</i> <i>What physical principle is Bernoulli's Equation really an expression of?</i>	Bernoulli's Equation: $P + \underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \text{constant}$ (along a streamline, for an ideal, incompressible, non-viscous fluid). Bernoulli's Equation is really a statement of $\underline{\hspace{2cm}}$ applied to a moving fluid — pressure energy, kinetic energy, and gravitational potential energy per unit volume must add up to the same total everywhere along the flow. Practical consequence: where a fluid speeds up (v increases), its pressure must $\underline{\hspace{2cm}}$, all else being equal.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

12.3 The Most General Applications of Bernoulli's Equation

CUE / KEY QUESTIONS <i>What does Bernoulli's Equation reduce to for a fluid at rest?</i> <i>Name two real-world applications of Bernoulli's Equation.</i>	For a fluid at rest ($v = 0$ everywhere), Bernoulli's Equation reduces to $P + \rho gh = \text{constant}$, which is just the fluid _____ pressure-with-depth relationship from Chapter 11. Applications: _____
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SUMMARY — in your own words, summarize this section in 2–3 sentences: _____ _____ _____

Chapter 16: Oscillatory Motion

16.2 Period and Frequency in Oscillations

CUE / KEY QUESTIONS <i>Define period T and frequency f.</i> <i>Equation relating T and f?</i> <i>SI unit of frequency?</i>	Period (T) is the time for one complete _____ (oscillation), measured in _____. Frequency (f) is the number of cycles per unit time: $f = \frac{1}{T}$, SI unit: _____, where $1 \text{ Hz} = 1 \text{ cycle/s}$.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

16.3 Simple Harmonic Motion: A Special Periodic Motion

CUE / KEY QUESTIONS <i>Define simple harmonic motion (SHM).</i> <i>What restoring-force condition defines SHM?</i> <i>Period of a mass on a spring?</i>	Simple Harmonic Motion (SHM) is oscillatory motion where the _____ is directly proportional to the displacement from equilibrium, and points _____ to that displacement. Condition for SHM (Hooke's Law form): $F = -kx$ Period of a mass m on a spring of constant k : $T = 2\pi\sqrt{m/k}$ — notice the period does NOT depend on amplitude.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

16.4 The Simple Pendulum

CUE / KEY QUESTIONS <i>Period of a simple pendulum (small-angle approximation)?</i> <i>What does the period of a simple pendulum depend on — and NOT depend on?</i>	Period of a simple pendulum (small angles, length L): $T =$ _____ The period depends only on the pendulum's _____ and the local _____. It does _____ depend on the _____ of the bob or (for small angles) on the _____ of the swing.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

16.6 Uniform Circular Motion and Simple Harmonic Motion

CUE / KEY QUESTIONS <i>How is SHM related to uniform circular motion?</i> <i>What does the angular frequency ω of SHM correspond to in the circular-motion picture?</i>	SHM can be visualized as the _____ of an object moving in uniform circular motion, projected onto one diameter of the circle. The angular frequency ω used in SHM equations corresponds directly to the _____ of the equivalent circular motion: $\omega =$ _____ = _____
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

Waves and Sound

16.9 Waves

CUE / KEY QUESTIONS <i>Define a wave.</i> <i>Difference between transverse and longitudinal waves?</i> <i>Wave speed equation relating v, f, and λ?</i>	A wave is a disturbance that moves _____ from one place to another without transporting _____ along with it. In a _____ wave, the medium moves perpendicular to the direction the wave travels (e.g., a wave on a string). In a _____ wave, the medium moves parallel to (along) the direction of travel (e.g., sound). Wave speed: $v = \frac{\lambda}{T}$, where λ (wavelength) is the distance between two identical points on the wave (e.g., crest to crest).
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

17.1 Sound

CUE / KEY QUESTIONS <i>What kind of wave is sound?</i> <i>What medium(s) can sound travel through?</i>	Sound is a _____ wave — a series of compressions and rarefactions traveling through a medium. Sound requires a physical medium to travel through — it can move through _____, but NOT through a vacuum.
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

17.2 Speed of Sound, Frequency, and Wavelength

CUE / KEY QUESTIONS <i>What does the speed of sound depend on?</i> <i>Approximate speed of sound in air at 20°C?</i>	The speed of sound depends on the properties of the _____ it travels through — mainly its _____ and _____ — and generally increases from gases to liquids to solids. Speed of sound in air at 20°C: $v \approx$ _____ Relationship (as with all waves): $v =$ _____
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

17.3 Sound Intensity and Sound Level

CUE / KEY QUESTIONS <i>Define sound intensity. Give the equation and SI unit.</i> <i>Equation for sound level (decibels)?</i> <i>Reference intensity I_0?</i>	Sound intensity (I) is power per unit area: $I =$ _____, SI unit: _____ Sound level (β), in decibels: $\beta =$ _____ dB Reference intensity: $I_0 =$ _____ (roughly the quietest sound a human can hear).
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

17.4 Doppler Effect and Sonic Booms

CUE / KEY QUESTIONS <i>Define the Doppler Effect.</i> <i>What happens to observed frequency when the source approaches vs. recedes?</i>	The Doppler Effect is the change in observed _____ (and wavelength) of a wave caused by _____ between the source and the observer. When the source moves toward the observer (or vice versa), the observed frequency _____ (pitch sounds higher). When the source moves away, the observed frequency _____ (pitch sounds lower).
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SUMMARY — in your own words, summarize this section in 2–3 sentences:

17.5 Sound Interference and Resonance: Standing Waves in Air Columns

CUE / KEY QUESTIONS <i>Define a standing wave.</i> <i>Resonant frequencies for a tube open at both ends?</i> <i>Resonant frequencies for a tube closed at one end?</i>	A standing wave is formed by the _____ of two waves of the same frequency traveling in _____, creating fixed points (nodes) that never move. Tube open at both ends (length L): resonant frequencies $f_n = \text{_____}$, $n = 1, 2, 3, \dots$ (all harmonics allowed). Tube closed at one end (length L): resonant frequencies $f_n = \text{_____}$, $n = 1, 3, 5, \dots$ (_____ harmonics allowed).
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SUMMARY — in your own words, summarize this section in 2–3 sentences:
